Web Engineering & Development (SWE 363)

Basics of HTML (Hypertext Markup Language)

Announcements 📣

- Project Assignment #1 is due this Saturday at 11:59 PM
- Project teams should be consist of 3-4 members
- For the project ideas, develop real-world apps that solve problems, benefit society, and generate revenue.

In today's Lecture:

- HTML Document structure
- Basic HTML Tags
- HTML Lists
- HTML Tables
- HTML Images

HTML

What?

HTML

- Hyper Text: Text with links to other documents or resources.
- Markup: Defines the structure and meaning of different parts of the document.
- Language: A set of rules for writing HTML code properly.

So what does CSS & JS do?

HTML Elements

```
<p>Hello</p>
```

HTML Elements

```
<p>Hello</p>
```

Examples include: `<h1>`, `<h2>`, `<h3>`, `<h4>`, `<h5>`, `<h6>`, `<ul>`, `<ol>`, `<li>`, `<table>`, `<tr>`, `<td>` etc.

HTML Attribute

```
<a href="https://google.com">Link</a>
```

HTML Attribute

```
<a href="https://google.com">Link</a>
```

Examples include: `href` , `src` , `alt` , `class` , `id` , `style` etc.

A Simple HTML Page

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <style>
      body { background-color: lightblue; }
    </style>
  </head>
  <body>
    Hello, world!
    <script>
      console.log('Hello from JavaScript!');
    </script>
  </body>
</html>
```

Let's Build a Simple HTML

HTML Elements reference

https://developer.mozilla.org/en-US/docs/Web/HTML

https://www.w3schools.com/html/default.asp

HTML common text elements

- Paragraph `<p>` : Used to define a paragraph of text
- Headings `<h1>` to `<h6>` : Used to define headings of different levels
- Formatting `<em> <i> <b> <mark>` : Used to format text

```
Activity: write the html code for the following:
```

SWE 363

Introduction to HTML

A course where we will learn the basics of HTML, CSS & JS.

HTML Lists

There are two types of lists: Ordered `<ol>` and Unordeered `<ul>`

```
<!-- Example of a list -->
<ul>
  <li>First item</li>
  <li>Second item</li>
  <li>Third item</li>
</ul>
```

How can we make a nested list?

HTML Tables

Consists of rows `<tr>` and columns `<td>`

```
<table>
  <tr>
    <td>Row 1, Cell 1</td>
    <td>Row 1, Cell 2</td>
    <td>Row 1, Cell 3</td>
  </tr>
</table>
```

Tables can get complex ...

```
<table>
  <thead>
    <tr>
      <th>Header 1</th>
      <th>Header 2</th>
      <th>Header 3</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td rowspan="2">Row 1, Cell 1 (spans 2 rows)</td>
      <td>Row 1, Cell 2</td>
      <td>Row 1, Cell 3</td>
    </tr>
    <tr>
      <td>Row 3, Cell 1</td>
      <td colspan="2">Row 3, Cell 2 (spans 2 columns)</td>
    </tr>
  </tbody>
  <tfoot>
    <tr>
      <td colspan="3">Footer (spans all columns)</td>
    </tr>
  </tfoot>
</table>
```

HTML Images

Images are *void* elements, they don't have a closing tag.

```
<img src="link_to_image.jpg" alt="image description">
```

Void elements

Void elements are elements that don't have a closing tag.

Examples: `<img>`, `<br>`, `<hr>`, `<input>`, `<link>`, `<meta>`, `<area>`, `<base>`, `<col>`, `<embed>`, `<hr>`, `<img>`, `<input>`, `<link>`, `<meta>`, `<param>`, `<source>`, `<track>`, `<wbr>`

Let's build a rich html page

Classroom50 Assignment: **2.1 HTML Basics**

Classroom's Repository Push Issue:

Set Your Git Username and Email to Match GitHub

```
git config --add --global user.name "Your GitHub Name"
git config --add --global user.email "your_email@kfupm.edu.sa"
```

Why This Matters?

GitHub links commits to your account using the email in your Git config. If the email doesn’t match any email in your GitHub account, your commits won’t be linked to your profile

Next Class

- Links
- Containers
- Forms